Qalibrated Systems Limited

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| Approved By | Technical Manager |

# CALIBRATION PROCEDURE

Electromechanical & Mechanical Manometers / Pressure Gauges — per EURAMET Calibration Guide No. 17

## 1. Scope & Application

This procedure applies to the calibration of electromechanical and mechanical manometers, pressure gauges, and digital pressure calibrators measuring absolute, gauge, and differential pressure, per EURAMET Calibration Guide No. 17, "Calibration of Electromechanical and Mechanical Manometers", covering Bourdon tube gauges and digital pressure indicators.

## 2. Reference Standards & Traceability

| STANDARD | DESCRIPTION | TRACEABILITY |
|---|---|---|
| Reference Pressure Standard / Deadweight Tester | Covers the full range of the instrument under test, with uncertainty appropriate to the target test uncertainty ratio | National metrology institute, ≲ 24 months |
| Pressure Comparator / Test Bench | Generates and controls test pressure to the required stability | Verified against reference standard |
| Reference Thermometer | Ambient and instrument temperature (for instruments sensitive to temperature) | Accredited lab, ≲ 24 months |

## 3. Environmental Conditions

| PARAMETER | REQUIREMENT |
|---|---|
| Thermal Equilibrium | Instrument and reference standard allowed to reach thermal equilibrium with the test environment before calibration |
| Mounting Position | Instrument calibrated in its normal operating orientation, as installed in service |
| Pressure Medium | Appropriate medium (gas or hydraulic fluid) per the instrument's intended use and the reference standard's specification |

## 4. Procedure

1. **Pre-Test Inspection.** Inspect the instrument for damage, leaks, and correct zero indication at ambient (atmospheric) pressure before pressurising.
2. **Select Procedure Tier.** Select Basic (6-point) or Standard (11-point) calibration procedure based on the target measurement uncertainty for the instrument.

   *Per cg-17: the Standard procedure is used where the expanded uncertainty (k=2) target is between 0.05% and 0.2% of full scale; the Basic procedure uses 6 points to evaluate hysteresis and repeatability.*
3. **Increasing Pressure Run.** Apply pressure in ascending steps to each selected test point, allowing stabilisation, and record the indication of the instrument under test against the reference standard at each point.
4. **Decreasing Pressure Run.** Repeat the test points in descending order to evaluate hysteresis.
5. **Repeat Cycles.** Repeat the full increasing/decreasing cycle a minimum of 2 additional times to establish repeatability.
6. **Zero Check.** Confirm the instrument returns to zero (within tolerance) at the end of the test sequence.

## 5. Uncertainty & Calculation

The combined standard uncertainty includes: the calibration uncertainty of the reference pressure standard; the repeatability of readings across cycles; the hysteresis observed between increasing and decreasing runs; the resolution of the instrument under test; and any temperature sensitivity correction. These are combined per the GUM method to give the expanded uncertainty (k=2) of the error of indication at each test point.

## 6. Acceptance Criteria

The instrument passes calibration where the error of indication at each test point, together with its uncertainty, falls within the manufacturer's accuracy class specification or the client's stated tolerance requirement.

| Prepared By | Reviewed By | Approved By |
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| Technical Manager | Quality Manager | Managing Director |